THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

December 15, 2014 TIME: 9:30 am – 12:30 pm

I. Answer ALL questions. Section A carries 60 marks, and Section B carries 40 marks.

II. Atomic weights: H 1; C 12; F 19; O 16; Na 23; Mg 24.3; P 31; S 32; Cl 35.5; K 39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (20 multiple choice questions with 3 marks each)

1. Sulfur dioxide (SO_2) concentration in a city reaches 2 mg/m^3 at 25 °C. Estimate its concentration in ppmv. (a) 0.4 ppmv, (b) 0.76 ppmv, (c) 1 ppmv, (d) 2 ppmv, or (e) 4 ppmv.
2. A factory has five identical machines. If each machine operates at 76 dB, what will be the combined noise level when all the machines operate together? (a) 76 dB, (b) 78 dB, (c) 79 dB, (d) 81 dB, (e) 83 dB, (f) 86 dB, or (g) 228 dB?
3. The initial mass of a radioactive isotope Pb-214 was 1 mg. At t = 3 hours, only 0.0095 mg of Pb-214 was left. What is the amount of Pb-214 left at t = 30 minutes? (a) 0.14 g, (b) 0.16 g, (c) 0.25 g, (d) 0.46 g, (e) 0.64 g, or (f) 0.80 g.
4. Which of the following process is suitable for hardness removal? (a) Air stripping, (b) ion exchange, (c) depth filtration, or (d) ultrafiltration.
5. Which of the following disinfectants is most suitable for secondary disinfection? (a) UV, (b) ozone, (c) chlorine dioxide, or (d) chloramine.
6. Hypochlorous acid (HOCl) is a much stronger disinfectant than its conjugate base hypochlorite (OCl^-). The pKa value of HOCl is 7.6. Determine [HOCl]/[OCl^-] at pH 6. (a) 100, (b) 40, (c) 16, (d) 1.6, or (e) 0.025.
7. What is the nitrogenous oxygen demand of a wastewater sample containing 100 mg/L $C_6H_5NO_2$? (a) 14 mg/L, (b) 42 mg/L, (c) 52 mg/L, (d) 70 mg/L, or (e) 100 mg/L.
8. Which type of membrane is most suitable for virus removal? (a) RO, (b) NF, (c) UF, or (d) MF.
9. Which biological treatment belongs to anaerobic treatment? (a) Activated sludge, (b) aerated lagoon, (c) trickling filter, (d) sludge digestion, or (e) rotating biological contactor.
10. Two identical disinfection contactors are connected in series. If each contactor is able to achieve a 3-log inactivation of target pathogens, which is the overall inactivation efficiency? (a) 1.5-log, (b) 3-log, (c) 90%, (d) 99.5%, (e) 99.999%, or (f) none of the above.
11. A groundwater has the following properties: pH 7.5, [HCO_3^-] = 50 mg/L, and [Ca^{2+}] = 50 mg/L. Determine its carbonate hardness as $CaCO_3$(s). (a) 41 mg/L, (b) 50 mg/L, (c) 100 mg/L, (d) 125 mg/L, or (e) none of the above.
12. Which of the following statement is correct with respect to dissolved oxygen (DO)? (a) Solubility of oxygen increases with temperature, (b) solubility of oxygen increases with salinity, (c) solubility of oxygen increases with elevation, (d) polluted river water typically has higher DO in summer compared to winter, or (e) none of the above.
13. Which of the following air pollutants is not a primary pollutant? (a) CO, (b) ground-level ozone, (c) NO_2, (d) SO_2, or (e) none of the above.
14. Which of the following statements is not correct with respect to NO_x and SO_x? (a) NO_x refers to NO and NO_2, (b) SO_x refers to SO_2 and SO_3, (c) both NO_x and SO_x can lead to acid rain problems, (d) both NO_x and SO_x are largely due to vehicle emissions, or (e) none of the above.
15. Which process is not suitable for removing particles from power plant flue gas? (a) Electrostatic precipitator, (b) cyclone, (c) baghouse, (d) three-way catalytic converter, or (e) gravity settler.
16. In a biological nutrient removal process, NO_3^- is converted to N_2 gas. The above phenomenon is known as (a) eutrophication, (b) nitrification, (c) denitrification, (d) nutrient precipitation, or (e) none of the above.

17. As the altitude increases, the air temperature decreases from 22°C at the ground level to 15°C at 500 m above the ground level. The temperature is constant (12°C) at altitude > 500 m. The air is dry so that the dry adiabatic lapse rate is applicable. Determine the mixing depth. (a) 200 m, (b) 300 m, (c) 400 m, (d) 500 m, (e) 600 m, or (f) 700 m.
18. The box model is commonly used to predict the indoor air quality. Which of the following conditions will lead to improved indoor air quality? (a) Increasing pollutant concentration in the ambient air, (b) increasing the source strength, (c) larger indoor air volume (i.e., a bigger box), or (d) larger infiltration rate.
19. Which of the following compounds does not have significant ozone depletion potential? (a) Halon-1211, (b) $CFCl_3$, (c) CF_2Cl_2, (d) HCFC-22, or (e) C_2F_6?
20. Which of the following effects is not a direct result of global climate change? (a) Sea level rise, (b) increased UV exposure, (c) global average temperature rise, (d) thermal stratification, (e) more frequent extreme weather conditions, or (f) potential change in ocean circulation pattern.

Section B (Two questions with 20 marks each)

21. (a) A city gets its supply of drinking water from a nearby river. Propose a typical drinking water treatment scheme. Briefly state the purpose of each treatment process.

(10 marks)

(b) Ocean acidification is a phenomenon caused by increased carbon dioxide concentration in the atmosphere. Briefly explain the phenomenon and its effect on ocean ecology.

(5 marks)

(c) A wastewater has a BOD_5 of 250 mg/L at 20°C. The rate constant at 20°C is 0.27 day^{-1}. What is the BOD_5 at 15°C? Assume a temperature correction factor of 1.047 in your calculation.

(5 marks)

22. (a) Phosphorus (P) can enter into a lake from both point sources (such as wastewater discharge) and non-point sources (such as surface runoff). An important mechanism for P removal from the lake water is sedimentation. In addition, P can leave the lake through lake outflows. Based on the above information, develop a mass balance equation for P in the lake water. Clearly define all the variables in the equation and state your assumption(s). How does the particle size of P-containing particles affect the concentration of P in the lake water?

(10 marks)

(b) What is eutrophication and how is eutrophication related to P?

(5 marks)

(c) For the lake in Part (a), its phosphorous concentration needs to be reduced to prevent severe eutrophication problem. What method(s) would you propose to reduce the P concentration?

(5 marks)

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